

LLM Training as per MLPerf benchmarks, and Rivera was bullish about Intel's road ahead in this competitive journey.

"We think from a price performance perspective, our offering is strong now and gets even better later this year," she said, hinting at a major software upgrade to enhance Gaudi2's performance with respect to the MLPerf benchmark that's expected to go public in November 2023. "Next year we will have Gaudi3 in the market, which will be competing very effectively with NVIDIA's H100 and their next generation chips, so we'll see. Our projections look very good, we have been catching up and doing pretty well with our portfolio, we're priced very aggressively, and there's a lot of room in the market for choice. Customers want choice and alternatives, and we absolutely want to be a competitor and alternative to the biggest player in the AI market. It's going to be what we do, not what we say," Rivera stated.

"We'll compete on the merits of our products, our approach is open choice and we don't believe in locking people in. We are big investors and leaders in building ecosystems," Rivera reiterated, saying with the continuum of AI use cases, Intel knows there are very broad and varied requirements for AI workloads. Which means one size fits all approach doesn't work, something that even Intel's competitors are realising, according to Rivera.

"Even our competitors are trying to build portfolios that have more than one architecture," she said. "You see GPU companies building CPUs, you see CPU companies acquiring FPGA assets, and the unifying layer for all of that is in the software," stated Rivera. Over 80-per cent of all AI development happens at the framework level and above, she suggested.

For instance, a biologist doing genome sequencing, meteorologist working on weather models or chemical engineer doing oil and gas research doesn't care about what's happening at an AI chip's kernel or hardware driver level – they only get involved at Python, or some other AI framework level, and



largely want to work up the software stack. "And the key for us at Intel is to make sure we're enabling those developers to get access to our innovations through those standard, high level languages and frameworks, and lowering their barrier to entry and easing their market participation as far as AI workloads and deployments are concerned," according to Rivera.

Talking about the core data center business and refresh cycles of Intel's ecosystem partners, Rivera highlighted how the industry's become more focused on driving efficiency and increasing overall infrastructure utilisation.

"One of the interesting things that happened during the pandemic is the supply chain constraints, and we saw customers lean into more utilisation of the infrastructure they have," Rivera recollected. Refresh cycles were dictated by which workloads were in demand. "If you're talking about AI, networking, security, customers demanded the latest innovations and cutting edge solutions without wasting any time, but if you look at a lot of the web-tier kind of traffic – like office applications, accounting systems, ERP systems – there we saw customers very focused on increasing utilisation of infrastructure resources they already have," mentioned Rivera. The same behaviour was observed in Indian data center customers as well.

According to Intel, a lot of India's data center infrastructure optimization is driven by power constraints. "We saw customers optimise around their data center power budget, for instance having power dedicated to leading edge applications and infrastructure and sort of cap some of the power being spent on more entry-level or mainstream applications," she said.

India isn't just an important market for Intel but the company's committed to investing in the country, as it always has in the past, suggested Rivera. "From an R&D perspective, Intel has been here for decades and invested almost \$9 billion in ramping up Intel India's tech capabilities over the years with a workforce that spans over 14,000 so far. Not just that, but we see all the investments that the government is making in digital infrastructure, so the opportunity to build more data centers, enterprise solutions, software ecosystem solutions and services is very exciting," Rivera underscored in no uncertain terms.

In just her division within Intel (pertaining to data center and AI), there are over 3,000 employees dedicated within India. "They are working on our CPUs, on FPGAs, our GPUs and AI accelerators. Everything we do under our data center portfolio, from a hardware, silicon development perspective, the team here in India is critical for all of those programs," mentioned Rivera. In fact, the Gaudi3 chip, which is coming to market in 2024, Intel India's silicon development team in Bengaluru is responsible for actually delivering that product together with Intel Israel's team.

"We are very proud of the capabilities of what the team has done and the quality, execution discipline and the competitiveness of the portfolio being delivered is excellent," Rivera emphasised. 

